# *Arturo Arellano Arias*

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## *Education*

**Ph.D. degree in Mathematics** — *McGill University*
*GPA: 4.0* — *September 2021 - present*
*Thesis project: Shape theorems for periodic branching diffusions*
*Advisors: Jessica Lin and Louigi Addario-Berry*

**Master's degree in Applied Mathematics** — *CIMAT, Mexico*
*GPA: 3.8* — *August 2019 - August 2021*
*Thesis: Percolación, el modelo de Ising y SLE*
*Advisor: Juan Carlos Pardo*

**Bachelor's degree in Mathematics** — *Universidad de Guanajuato, Mexico*
*GPA: 3.5* — *August 2014 - August 2019*
*Thesis:Una mirada al Teorema Ergódico Subaditivo*
*Advisor: Renato Iturriaga*

## *Awards and fellowships*

**FRQN Doctoral Reasearch Scholarship** — *May 2023 - December 2026*
*Fonds de recherche du Québec – Nature et technologies*

**MITACS Globalink Research Award** — *April 2023*
*MITACS*

**CONACYT National Graduate Scholarship** — *August 2019 - August 2021*
*Consejo Nacional de Ciencia y Tecnología, Mexico*

**CIMAT Undergraduate Scholarship for Excellence** — *August 2014 - August 2019*
*CIMAT*

**Gold medal Mexican Mathematics Olympiad** — *November 2013*
*Sociedad Matemática Mexicana*

## *Publcations and preprints*

A. A. A. **"A shape theorem for periodic branching diffusion"** — *August 2026*
arXiv:2608.17031

J. Amorim, A. A. A, M. Jara. **"Scaling limit of Smoluchowski particles"** — *March 2026*
arXiv:2603.25868

L. Addario-Berry, A. A. A, J. Lin. **"A shape theorem for BBM in a periodic environment"** — *July 2025*
arXiv:2507.10515

## *Ongoing work*

A. A. A., H. Chang-Lara. **"A discrete scheme for a transmission problem combining a firs-order and a second-order operator"** — *In preparation*

## *Teaching experience*

**Instructor, MATH 203: Principles of Statistics** *Summer 2026*
*McGill University* *Montreal, Canada*
Introductory undergraduate statistics course for students across a wide range of programs: descriptive statistics, common distributions, confidence intervals and hypothesis tests for means, variances and proportions.

**Teaching Assistant, MATH 447: Introduction to Stochastic Processes.** *Winter 2026*
*McGill University* *Montreal, Canada*
Upper-level undergraduate course in stochastic processes: conditional expectation, generating functions, branching processes, random walks, Markov chains and the ergodic theorem, birth-death processes and queueing theory.

**Teaching Assistant, MATH 323: Probability.** *Fall 2024*
*McGill University* *Montreal, Canada*
Undergraduate probability course: conditional probability and Bayes' theorem, combinatorial probability, discrete and continuous random variables and distributions, independence, probability inequalities, law of large numbers and central limit theorem..

**Teaching Assistant, MATH 315: Ordinary Differential Equations.** *Fall 2022*
*McGill University* *Montreal, Canada*
Undergraduate course in ordinary differential equations: first-order equations and elementary numerical methods, linear differential equations, Laplace transforms and series solutions.

**Teaching Assistant Elements of Geometry, Calculus I, II and III** *2018-2021*
*Universidad de Guanajuato* *Guanajuato, Mexico*
First- and second-year courses for mathematics majors: Euclidean geometry, single- and multivariable differential and integral calculus. Taught in Spanish.

## *Mentoring*

**Directed Reading Program** *Winter 2023, Winter 2024*
*Graduate Mentor* *McGill University*
Mentored two undergraduate students in semester-long reading projects on Monte Carlo Simulations, and Stochastic processes, through weekly one-on-one meetings.
Guided each student in writing an expository paper and preparing a final presentation.

**Coach Mexican Mathematics Olympiads** *2013 - 2021*
*Coach and Member of the State Organizing Committee* *Guanajuato and Campeche, Mexico*
Organized the state olympiad at every stage, from the initial school-level rounds to the final selection of the six students representing the state at the national competition.
Coached students in olympiad problem solving through regular training sessions.

## *Talks and Posters*

**A shape theorem for the convex hull of $d$-dimensional branching Brownian motion in periodic environments**

- BIRS Workshop on Reaction-Diffusion, Branching Processes, and Biology, Banff *May 2025*
- SIAM Student Chapter, Georgia Tech *April 2025*
- CMS Winter Meeting *December 2024*

**Branching Brownian motion in a periodic environments (Poster)**

- Conference on New Developments in Probability, CRM, Montreal *September 2024*

**A shape theorem for multidimensional branching Brownian motion**

- Students' Probability Seminar, IMPA, Rio de Janeiro *May 2024*

**A numerical method for the solution of a transmission problem**

- Students' Seminar in Applied Mathematics, McGill University *May 2023*

**Un algoritmo para aproximar la mejor estrategia en un juego estocástico**

- Junior Seminar, Universidad de Guanajuato *November 2022*

## *Technical skills*

**Programming Languages/Tools** Python, R, GitHub, LaTeX

## *Language proficiencies*

**English** Fluent
**Spanish** Native